Chapter 2 : Complex numbers

Tags	Courses

> 9th session: 22/10/2024

Operations on $(\mathbb{R}^2)$

Let us consider the set $(\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) \mid x, y \in \mathbb{R}\})$.

We define two operations on this set as follows:

For all $(x, y), (x', y') \in \mathbb{R}^2)$:

1. **Addition**:

   $$(x, y) + (x', y') = (x + x', y + y')$$

2. **Multiplication**:

   $$(x, y) \times (x', y') = (xx' - yy', xy' + yx')$$

Properties of Addition

- The operation $(+)$ defines an **Abelian (commutative) group**.
- The identity element for addition is:

  $$(0, 0)$$

- For all $((x, y) \in \mathbb{R}^2)$, there exists $((x_1, y_1))$ such that:

  $$(x, y) + (x_1, y_1) = (0, 0)$$

  where:

  $$x_1 = -x, \quad y_1 = -y$$

Properties of Multiplication

Now, consider the second operation $(\times)$.

Let $(e = (e_1, e_2)$ be the identity element of $(\mathbb{R}^2)$ with respect to \(\times \).

For all $((x, y) \in \mathbb{R}^2)$:

$$(x, y) \times (e_1, e_2) = (e_1, e_2) \times (x, y) = (x, y)$$

From:

1. $xe_1 - ye_2 = x$
2. $xe_2 + ye_1 = y$

We deduce:

$$e_1 = 1, \quad e_2 = 0$$

Thus, the identity element for \(\times \) is:

$$e = (1, 0)$$

For all $((x, y) \in \mathbb{R}^2)$:

$$(x, y) \times (1, 0) = (1, 0) \times (x, y) = (x, y)$$

Inverse with Respect to Multiplication

Let $((x, y))$ be any element of $(\mathbb{R}^2)$, and consider $((x_1, y_1))$ as its inverse:

$$(x, y) \times (x_1, y_1) = (1, 0)$$

From:

1. $xx_1 - yy_1 = 1$
2. $xy_1 + x_1y = 0$

We find:

$$x_1 = \frac{x}{x^2+y^2}, \quad y_1 = -\frac{y}{x^2+y^2}$$

Associativity and Field Properties

Both $(+)$ and $(\times)$ are associative, and thus $(\mathbb{R}^2, +, \times))$ forms a **field**.

Complex Numbers and $(\mathbb{C})$

Define:

$$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}, \, i^2 = -1\}$$

Equivalently:

$$\mathbb{C} = \{x(1,0) + y(0,1) \mid x, y \in \mathbb{R}\} = \{(x, y) \mid x, y \in \mathbb{R}\}$$

Example

Evaluate:

$$(0,1) \times (0,1) = (-1,0)$$

This shows:

$$i \times i = (-1,0) = -(1,0) = -1$$

> ### Example: Evaluate
>
> $$1 + i + i^2 + \ldots + i^n = \frac{1 - i^{n+1}}{1 - i} = \begin{cases} 1 & if\ n = 4k \\ 1+i & if\ n = 4k+1 \\ i & if\ n = 4k+2 \\ 0 & if\ n = 4k+3 \end{cases}$$

Remarks on Complex Numbers

1. For $z = a + ib$, the **real part** is:

   $$\text{Re}(z) = a$$

   The **imaginary part** is:

   $$\text{Im}(z) = b$$

2. If $(b = 0)$, (z) is a **real number**.
3. If $(a = 0)$, z is a **purely imaginary number**.

For $(z_1 = a_1 + ib_1)$ and $(z_2 = a_2 + ib_2)$:

$$z_1 = z_2 \iff a_1 = a_2 \text{ and } b_1 = b_2$$

Definitions

1. **Absolute Value (Modulus)**:

   For $(z = a + ib)$:

$$|z| = \sqrt{a^2 + b^2}$$

2. **Argument**:

   For $(z = a + ib)$:

   $\cos(\arg z) = \frac{a}{\sqrt{a^2+b^2}}, \quad \sin(\arg z) = \frac{b}{\sqrt{a^2+b^2}}$

3. **Principal Argument**:

   The principal argument satisfies:

   $-\pi < \mathrm{Arg}(z) \le \pi$

The following properties hold

1) $\overline{z_1 \pm z_2} = \overline{z_1} \pm \overline{z_2}$,

2) $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$,

3) $\overline{\dfrac{z_1}{z_2}} = \dfrac{\overline{z_1}}{\overline{z_2}}$.

4) $|z| \geqslant 0$ et $|z| = 0 \Leftrightarrow z = 0$,

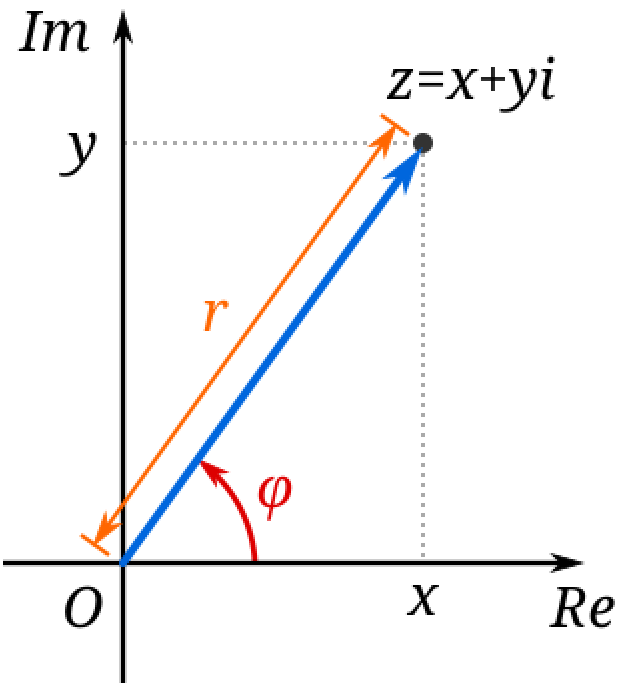
5) $|z_1 z_2| = |z_1|\,|z_2|$

6) $||z_1| - |z_2|| \leqslant |z_1 - z_2| \leqslant |z_1| + |z_2|$

7) $|Rez| \leqslant |z|\,,\ |Imz| \leqslant |z|$.

8) $\dfrac{1}{z} = \dfrac{\overline{z}}{|z|}$.

Complex plane , and polan coordinates



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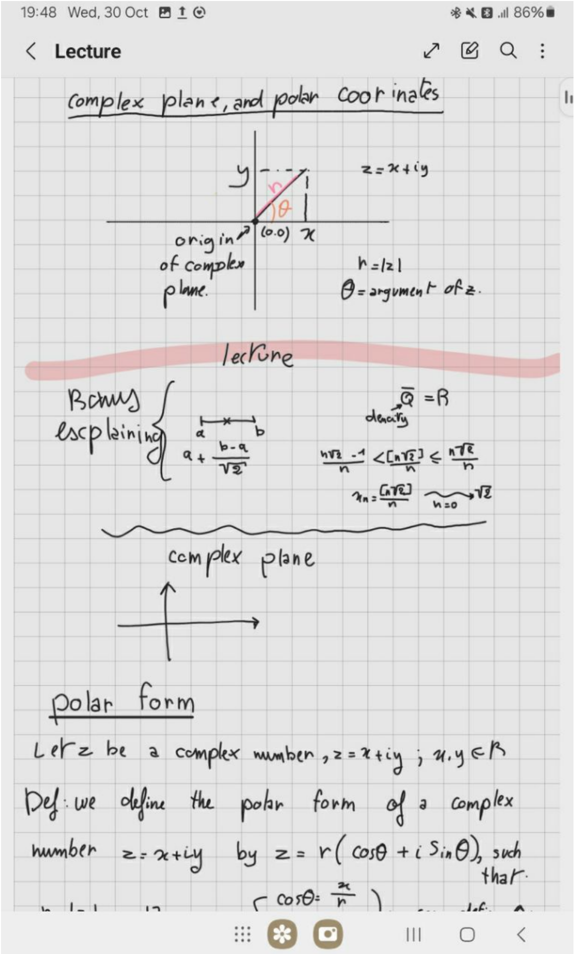

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Lecture

complex plane, and polar coorinates

y

$z = x + iy$

r

θ

origin of complex plane. (0.0) x

$r = |z|$

θ = argument of z.

lecture

Bawy escplaining

a b

$a + \frac{b-a}{\sqrt{2}}$

$\bar{Q} = \mathbb{R}$

density

$\frac{n\sqrt{2} - 1}{n} < \frac{[n\sqrt{2}]}{n} \leq \frac{n\sqrt{2}}{n}$

$x_n = \frac{[n\sqrt{2}]}{n} \xrightarrow[n=0]{} \sqrt{2}$

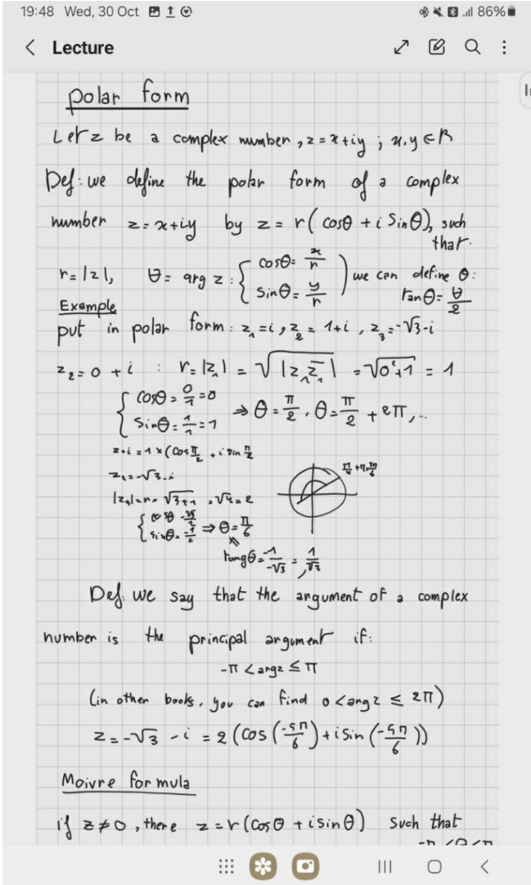
complex plane

polar form

Let z be a complex number, $z = x + iy$; $x, y \in \mathbb{R}$

Def: we define the polar form of a complex number $z = x + iy$ by $z = r(\cos\theta + i\sin\theta)$, such that.

$\cos\theta = \frac{x}{r}$



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Lecture

polar form

Let z be a complex number, $z = x + iy$; $x, y \in \mathbb{R}$

Def: we define the polar form of a complex number $z = x + iy$ by $z = r(\cos\theta + i\sin\theta)$, such that.

$r = |z|$, $\theta = \arg z : \begin{cases} \cos\theta = \frac{x}{r} \\ \sin\theta = \frac{y}{r} \end{cases}$ we can define θ: $\tan\theta = \frac{\theta}{2}$

Example

put in polar form: $z_1 = i$, $z_2 = 1 + i$, $z_3 = -\sqrt{3} - i$

$z_2 = 0 + i$: $r = |z_1| = \sqrt{|z_1 \bar{z}_1|} = \sqrt{0^2 + 1} = 1$

$\begin{cases} \cos\theta = \frac{0}{1} = 0 \\ \sin\theta = \frac{1}{1} = 1 \end{cases} \Rightarrow \theta = \frac{\pi}{2}, \theta = \frac{\pi}{2} + 2\pi, \dots$

$z = i = 1 \times (\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$

$z_3 = -\sqrt{3} - i$

$|z_3| = r = \sqrt{3+1} = \sqrt{4} = 2$

$\begin{cases} \cos\theta = \frac{-\sqrt{3}}{2} \\ \sin\theta = \frac{-1}{2} \end{cases} \Rightarrow \theta = \frac{\pi}{6}$

$\tan\theta = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}}$

$\frac{\pi}{6} + \pi = \frac{7\pi}{6}$

Def: we say that the argument of a complex number is the principal argument if:

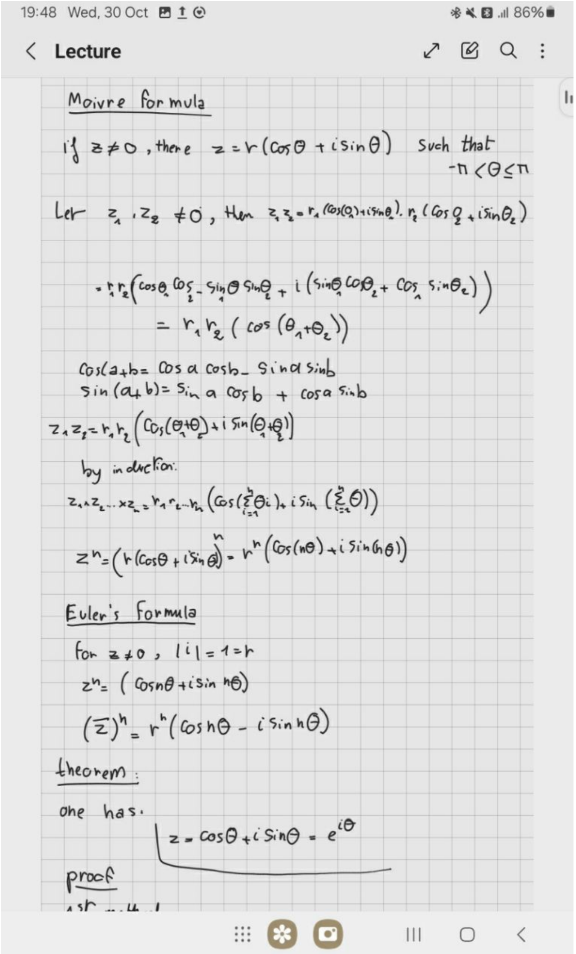
$-\pi < \arg z \leq \pi$

(in other books, you can find $0 < \arg z \leq 2\pi$)

$z = -\sqrt{3} - i = 2\left(\cos\left(\frac{-5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right)$

Moivre formula

if $z \neq 0$, there $z = r(\cos\theta + i\sin\theta)$ such that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Lecture

Moivre formula

if $z \neq 0$, there $z = r(\cos\theta + i\sin\theta)$ such that $-\pi < \theta \leq \pi$

Let $z_1, z_2 \neq 0$, then $z_1 z_2 = r_1(\cos(\theta_1) + i\sin\theta_1) \cdot r_2(\cos\theta_2 + i\sin\theta_2)$

$$= r_1 r_2\left(\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2 + i\left(\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2\right)\right)$$

$$= r_1 r_2 \left(\cos(\theta_1 + \theta_2)\right)$$

$\cos(a+b) = \cos a \cos b - \sin a \sin b$
$\sin(a+b) = \sin a \cos b + \cos a \sin b$

$$z_1 z_2 = r_1 r_2 \left(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)\right)$$

by induction:

$$z_1 \times z_2 \cdots \times z_n = r_1 r_2 \cdots r_n \left(\cos\left(\sum_{i=1}^{n} \theta_i\right) + i\sin\left(\sum_{i=1}^{n} \theta\right)\right)$$

$$z^n = \left(r(\cos\theta + i\sin\theta)\right)^n = r^n\left(\cos(n\theta) + i\sin(n\theta)\right)$$

Euler's Formula

for $z \neq 0$, $|i| = 1 = r$

$z^n = (\cos n\theta + i\sin n\theta)$

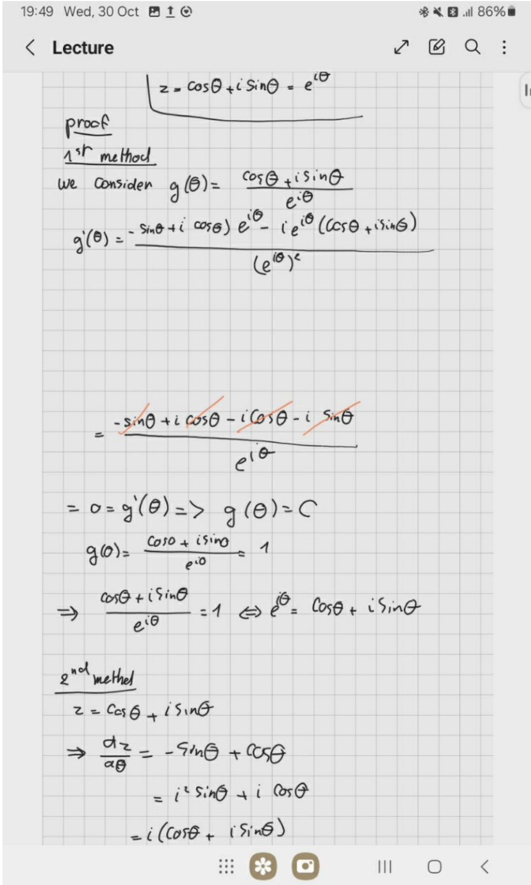
$(\bar{z})^n = r^n(\cos n\theta - i\sin n\theta)$

theorem:

one has.

$$z = \cos\theta + i\sin\theta = e^{i\theta}$$

proof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Lecture

$$z = \cos\theta + i\sin\theta = e^{i\theta}$$

proof

1st method

we consider $g(\theta) = \dfrac{\cos\theta + i\sin\theta}{e^{i\theta}}$

$$g'(\theta) = \frac{(-\sin\theta + i\cos\theta)e^{i\theta} - ie^{i\theta}(\cos\theta + i\sin\theta)}{(e^{i\theta})^2}$$

$$= \frac{\cancel{-\sin\theta} + \cancel{i\cos\theta} - \cancel{i\cos\theta} - \cancel{i\sin\theta}}{e^{i\theta}}$$

$$= 0 = g'(\theta) \Rightarrow g(\theta) = C$$

$$g(0) = \frac{\cos 0 + i\sin 0}{e^{i0}} = 1$$

$$\Rightarrow \frac{\cos\theta + i\sin\theta}{e^{i\theta}} = 1 \Leftrightarrow e^{i\theta} = \cos\theta + i\sin\theta$$

2nd method

$z = \cos\theta + i\sin\theta$

$$\Rightarrow \frac{dz}{d\theta} = -\sin\theta + \cos\theta$$

$$= i^2\sin\theta + i\cos\theta$$

$$= i(\cos\theta + i\sin\theta)$$